



Argand Diagrams · Worksheet 1

Core Pure 1 Chapter 2: Argand Diagrams. Answer all questions. Show your working clearly. Total: 81 marks. Time: 100 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 2. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Give arguments in radians, as principal arguments in the range $-\pi < \theta \leq \pi$. Give exact answers where possible, otherwise answers to 3 significant figures.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Modulus and argument	19	
B Modulus–argument form	18	
C Loci: circles, half-lines and perpendicular bisectors	18	
D Regions and combined loci	21	
Roar-level challenge	5	
Total	81	

Section A Modulus and argument

19 marks

1. Given that $z = 3 - 4i$, find $|z|$ and $\arg z$. [2]

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2. The complex numbers $z_1 = 2 + i$ and $z_2 = -1 + 3i$. [2]
Show z_1 , z_2 and $z_1 + z_2$ on a single Argand diagram.



3. Find the exact modulus and the exact argument of

[3]

(a) $-\sqrt{3} + i$,

(b) $-1 - i$.

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4. Given that $z = \frac{1 + 7i}{1 - i}$, express z in the form $a + bi$, and hence find $|z|$ and $\arg z$.

[4]

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5. The complex number z is given by $z = -2 + 2\sqrt{3}i$.

(a) Find the exact values of $|z|$ and $\arg z$. [2]

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(b) Express z^2 in the form $a + bi$, where a and b are exact. [2]

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(c) Find $|z^2|$ and $\arg(z^2)$. [2]

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(d) Show the points representing z and z^2 on a single Argand diagram. [2]

(Total for Question 5 is 8 marks)



Section B Modulus–argument form

18 marks

6. Express $4\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$ in the form $a + bi$, where a and b are exact. [2]

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7. Express $1 - i$ in the form $r(\cos \theta + i \sin \theta)$. [2]

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8. Express $-\sqrt{3} - 3i$ in the form $r(\cos \theta + i \sin \theta)$, giving r and θ exactly. [3]

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9. [4]

$$z_1 = 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right), \quad z_2 = 3\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$$

Express $z_1 z_2$ and $\frac{z_1}{z_2}$ in the form $r(\cos \theta + i \sin \theta)$.

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10.

$$z = \frac{1 + \sqrt{3}i}{1 - i}$$

- (a) Express $1 + \sqrt{3}i$ and $1 - i$ in the form $r(\cos \theta + i \sin \theta)$. [2]

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- (b) Hence express z in the form $r(\cos \theta + i \sin \theta)$. [2]

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- (c) By also expressing z in the form $a + bi$, show that $\cos \frac{7\pi}{12} = \frac{\sqrt{2} - \sqrt{6}}{4}$. [3]

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(Total for Question 10 is 7 marks)



Section C Loci: circles, half-lines and perpendicular bisectors

18 marks

11. Describe geometrically the locus of points z satisfying $|z - 3 + 2i| = 4$. [2]

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12. Sketch, on an Argand diagram, the locus of points z satisfying $\arg(z - 1) = \frac{\pi}{4}$. [2]

13. Find a Cartesian equation of the locus of points z satisfying $|z - 4| = |z + 2i|$. [3]

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14. The complex number z satisfies $|z + 1 - 2i| = 3$. [4]

(a) Find a Cartesian equation of the locus of z .

(b) Find the exact maximum value and the exact minimum value of $|z|$.

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15. (a) Sketch, on the same Argand diagram, the loci given by

[2]

$$|z - 3i| = 3 \quad \text{and} \quad \arg(z + 3) = \frac{\pi}{4}.$$

(b) Find a Cartesian equation of the half-line $\arg(z + 3) = \frac{\pi}{4}$.

[2]

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(c) Find, in the form $a + bi$ with a and b exact, the complex numbers that satisfy both equations.

[3]

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(Total for Question 15 is 7 marks)



Section D Regions and combined loci

21 marks

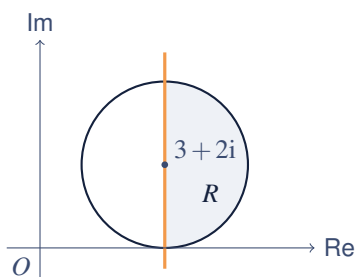
16. Shade, on an Argand diagram, the region satisfying $|z - 2| \leq 2$.

[2]



17. The shaded region R in the Argand diagram is bounded by part of a circle with centre $3 + 2i$ and radius 2, and part of the line $\operatorname{Re}(z) = 3$.

[3]



- (a) Write down two inequalities, in terms of z , that define R .
(b) Find the exact area of R .

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18. Show that the inequality $|z - 6| \leq |z - 2i|$ is equivalent to $y \leq 3x - 8$, where $z = x + yi$, and shade the region on an Argand diagram. [3]

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19. Shade, on an Argand diagram, the region satisfying both [4]

$$|z| \leq 4 \quad \text{and} \quad \frac{\pi}{6} \leq \arg z \leq \frac{2\pi}{3},$$

and find the exact area of this region.

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20. The region R is defined by

$$R = \{z : |z - 4 - 2i| \leq 2\} \cap \{z : |z - 4| \leq |z - 4i|\}.$$

(a) Show that the locus $|z - 4| = |z - 4i|$ has Cartesian equation $y = x$. [2]

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(b) Shade the region R on an Argand diagram. [3]

(c) Find the exact area of R . [4]

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(Total for Question 20 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

21. The complex number z satisfies $|z - 4i| = 2$. [5]
Find the greatest and least possible values of $\arg z$, and find, in the form $a + bi$, the value of z for which $\arg z$ is greatest.

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END OF WORKSHEET

Total: 81 marks